Exam practice

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Question 1
Answer saved
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Flag question

What is a React component?

Select one:

a. A CSS file that styles the entire application
b. A configuration file for the development server
c. A reusable piece of UI defined as a function
d. An HTML template loaded from an external file

Question 2
Answer saved
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Flag question

When should you use state in a React component?

Select one:

a. When you need to import external libraries
b. When the value changes and the UI should update
c. When the value never changes during runtime
d. When you need to pass data to a child component

Question 3
Answer saved
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Flag question

Which best describes JSX in React?

Select one:

a. A CSS preprocessor for styling React components
b. A testing framework for React applications
c. A separate programming language replacing JavaScript
d. A syntax extension allowing HTML-like code in JavaScript

Question 4
Answer saved
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Flag question

Why is asynchronous JavaScript important in web applications?

Select one:

a. It replaces the need for using promises and callbacks
b. It runs JavaScript code before the HTML page loads
c. It prevents blocking while waiting for slow operations
d. It runs multiple functions at the exact same time

Question 5

Not yet answered

Not graded

Flag question

Consider the following code:

```
1  const button = document.getElementById("myButton");
2  const display = document.getElementById("counter");
3  let count = 0;
4
5  button.addEventListener("click", () => {
6      count = count + 1;
7      display.textContent = "Clicks: " + count;
8  });
```

Based on the given code, answer the questions:

a) What does the code do when user clicks the button?
b) Why is the variable `count` defined with keyword `let`, but the `button` and `display` are defined with keyword `const`?

Question 6

Not yet answered

Marked out of 1

Flag question

What is the main purpose of using a CSS framework like Bootstrap?

Select one:

- a. It provides ready-made components for faster development
- b. It replaces the need for writing HTML code entirely
- c. It automatically makes websites accessible for all users
- d. It compiles CSS code into optimized JavaScript files

Question 7

Not yet answered

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Flag question

What does "cascading" refer to in Cascading Style Sheets?

Select one:

- a. CSS files are loaded and applied in the order they are linked
- b. Browser default styles override custom developer styles
- c. Multiple styles apply and conflicts resolve by rules
- d. Styles applied to a class override styles applied to an ID

Question 8

Not yet answered

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Flag question

Which CSS selector has the highest specificity value?

Select one:

- a. .item (class selector)
- b. * (universal selector)
- c. p (element selector)
- d. #main (id selector)

Question **9**

Not yet answered

Not graded

Flag question

Consider the following code:

```
function ProductCard(props) {
    return (
        <div className="card">
            <h2>{props.name}</h2>
            <p>Price: {props.price}€</p>
        </div>
    );
}

function App() {
    return (
        <div>
            <ProductCard name="Coffee" price={2.50} />
            <ProductCard name="Tea" price={2.00} />
        </div>
    );
}
```

Based on the given code, answer the questions:

a) Why is `ProductCard` defined as a separate component?
b) How does the `App` component use `ProductCard` component?
c) What JSX syntax is used in the code?

Question 10

Not yet answered

Marked out of 1

Flag question

What is a REST API used for in web development?

Select one:

- a. To handle browser events like clicks and form submissions
- b. To communicate between two systems over HTTP
- c. To provide reusable UI components across pages
- d. To convert JavaScript code into HTML content

Question 11

Not yet answered

Not graded

Flag question

Consider the following code:

```
1  form.addEventListener("submit", (event) => {
2      event.preventDefault();
3      const task = input.value;
4      const li = document.createElement('li');
5      li.textContent = task;
6      list.appendChild(li);
7      input.value = "";
8  });
```

Based on the code block, answer the questions:

a) What is this event listener for?
b) The code uses `event.preventDefault()`. Based on the context, what do you think happens if this line is removed?
c) What's the meaning of the row including `document.createElement("li")`?

Question 12

Not yet answered

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Flag question

Which HTML attribute is essential for making images accessible to screen reader users?

Select one:

- a. src
- b. width
- c. title
- d. alt

Question **13**

Not yet answered

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Flag question

When fetching data from an API in a React component, where should the fetch call typically be placed?

Select one:

- a. In a <script> tag placed directly in the JSX
- b. Directly in the component body without any wrapper
- c. Inside a useState hook that stores the fetched data
- d. Inside a useEffect hook that runs when needed

Question **14**

Not yet answered

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Flag question

What is the main difference between the Internet and the Web?

Select one:

- a. The Internet is the network infrastructure, the Web is an application on it
- b. The Web is the physical network, the Internet is the content
- c. They are the same technology with different names
- d. The Web is older and more fundamental than the Internet

Question **15**

Not yet answered

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Flag question

You want to group related HTML elements together as a container without adding any specific semantic meaning. Which tag is most appropriate?

Select one:

- a. <section>
- b.
- c. <article>
- d. <h1>

Question **16**

Not yet answered

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Flag question

You have finished editing a file and want to save the changes permanently to your local repository. Which command do you use?

Select one:

- a. git push
- b. git commit
- c. git status
- d. git add

Question 17

Not yet answered

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Flag question

A designer creates a website with high color contrast between text and background. Which WCAG principle does this support?

Select one:

- a. Understandable - content is clear
- b. Robust - content works on all devices
- c. Perceivable - content can be seen or heard
- d. Operable - content can be used by keyboard

Question 18

Not yet answered

Marked out of 1

Flag question

A user updates their existing profile information on a website. Which HTTP method is most appropriate for this action?

Select one:

- a. POST
- b. DELETE
- c. PUT
- d. GET

Question 19

Not yet answered

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Flag question

How do Git and GitHub relate to each other?

Select one:

- a. Git runs locally, GitHub automatically syncs versions to other developers
- b. GitHub is a required add-on that must be installed for Git to work
- c. Git runs locally on your computer, GitHub hosts repositories online
- d. Git is a command-line tool, GitHub is its graphical user interface

Question 20

Not yet answered

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Flag question

Which class targets medium devices (≥768px) in Bootstrap's grid system?

Select one:

- a. .col-md-4
- b. .col-xl-4
- c. .col-sm-4
- d. .col-lg-4

Question 21

Not yet answered

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Flag question

Which of these describes an event listener's role in JavaScript?

Select one:

- a. It selects HTML elements from the DOM
- b. It listens for changes in a variable and re-renders the DOM
- c. It listens to server messages and updates the page automatically
- d. It listens for user actions and runs a function

Question 22

Not yet answered

Not graded

Flag question

Consider the following code:

```
const button = document.getElementById('searchBtn');
button.addEventListener('click', async () => {
    const input = document.getElementById('search').value;
    try {
        const response = await fetch(`https://api.example.com/search?q=${input}`);
        const data = await response.json();
        console.log(data);
    } catch (error) {
        console.error('Error:', error);
    }
});
```

Based on the given code, answer the questions:

a) What does the code do, when a user clicks the button?
b) What is the purpose of the `async` keyword in this code? Why is it needed?

Question 23

Not yet answered

Marked out of 1

Flag question

You want to change the visible text of an HTML element from JavaScript. Which DOM method is most appropriate?

Select one:

- a. textContent
- b. getElementById
- c. createElement
- d. addEventListener

Question 24

Not yet answered

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Flag question

What does client-side routing in a Single Page Application do?

Select one:

- a. It changes the URL and view without reloading
- b. It stores the current component into React state
- c. It reloads the entire page when navigation happens
- d. It creates a new component for each URL automatically

Question 25

Not yet answered

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Flag question

What does the useState hook return in a React component?

Select one:

- a. An event listener attached to the component
- b. A single value that can be updated directly
- c. A promise that resolves with fetched data
- d. The current value and a function to update it

Question 26

Not yet answered

Not graded

Flag question

Consider the following code:

```
1  const numbers = [2, 5, 10, 15, 20];
2  const theseNumbers = numbers.filter(num => num > 10);
3  const thoseNumbers1 = theseNumbers.map(function(num) {
4      return num * 2;
5  });
6  const thoseNumbers2 = theseNumbers.map(num => num * 2);
```

Based on the given code, answer the questions:

a) Explain what the `filter`- and `map`-methods do in this code.
b) What does `num` represent in the callback functions of `filter` and `map`? Where does its value come from?
c) Explain what makes `thoseNumbers1` and `thoseNumbers2` equivalent. Why does the second version not need the `return` keyword?